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UNIVERSITY OF MORATUWA



EN 2160 Electronic Design Realization

Autonomous floor cleaning Robot

Project Report

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Team Fussion Runners

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1. Introduction, Problem Description and Justification

1.1 Background

The cleaning of floors in homes, boarding schools, office premises, and classrooms in Sri Lanka is a repetitive and laborious process that is currently done manually or with conventional vacuum cleaners that need to be operated continuously by humans. Robotic vacuum cleaners are commercially available all around the world; however, they have yet to gain popularity due to their prohibitive initial cost, poor adaptability within the region, and the fact that they are not designed to suit the typical local floor types such as tiled floors and cement floors.

Low-cost and locally available robotic cleaners provide basic sweeping capabilities without any obstacle detection and hence do not perform very well in terms of cleaning results.

1.2 Problem Statement

There is an obvious market gap between expensive imported robots which work well but cannot be afforded and are not supported locally, and low-priced robots which cannot be used due to their functional inefficiencies. The project aims to fill that gap by creating Leo, a cheap autonomous floor cleaner robot that uses LiDAR navigation, two-stage cleaning process (sweeping and vacuuming) and mobile app control implemented using a unique two board electronic design as opposed to standard development boards in order to reduce costs but preserve functionalities.

1.3 Survey Methodology

To justify the problem and measure interest from the market, prior to designing the product, we distributed a well-designed questionnaire through Google Forms to people living in households, people living in boarding homes, and office workers in our accessible network range. The data gathered included (i) their floor cleaning practices and the amount of time they spent in the house, (ii) their perceived usefulness of having a self-moving floor cleaner, and (iii) what kind of features they would like to see in the product. We got between 114 and 120 responses for each question (number of responses for each question is shown below each chart).

1.4 Survey Results

Time spent at home during the day

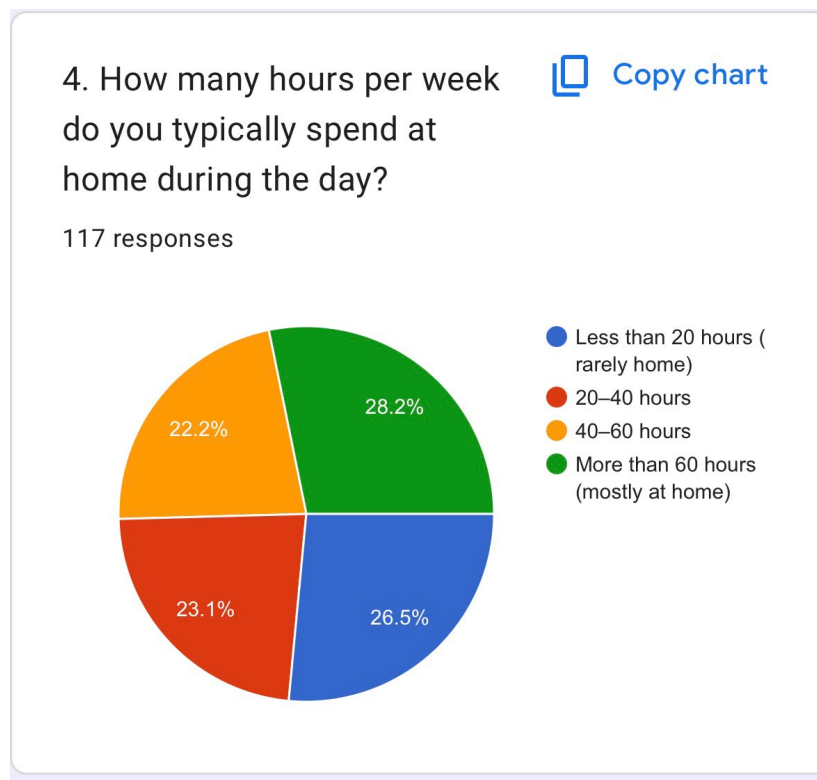


Figure 1.1: Reported hours per week typically spent at home during the day.

Respondents are fairly evenly spread across the four bands, with a slight lean toward the extremes: 26.5% are rarely home (under 20 hours/week) and 28.2% are mostly at home (over 60 hours/week), while the remaining 45.3% fall in between. Taken together, over half of respondents (50.4%) spend more than 40 hours a week at home. This confirms two distinct use cases the design must support: fully unattended autonomous operation for the large share of users who are frequently out, and simple, low-effort manual/app-triggered operation for users who are home most of the time and may prefer to start a cleaning cycle themselves.

Perceived usefulness of an automated floor-cleaning robot

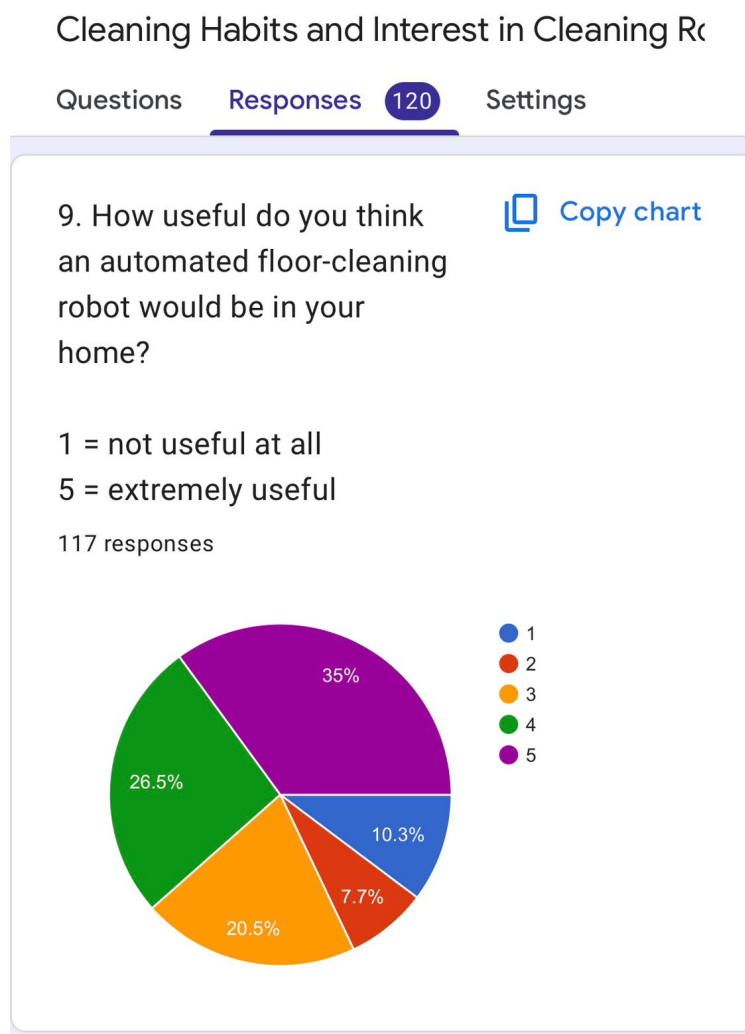


Figure 1.2: Perceived usefulness of an automated floor-cleaning robot at home, rated 1 (not useful at all) to 5 (extremely useful).

The response is strongly positive: 61.5% of respondents rated usefulness as 4 or 5, and 35% alone chose the top rating of 5. Only 18% (10.3% + 7.7%) rated the concept 1 or 2. This gives clear, survey-validated evidence of latent demand for the product and directly supports the decision to proceed with development.

Feature priorities

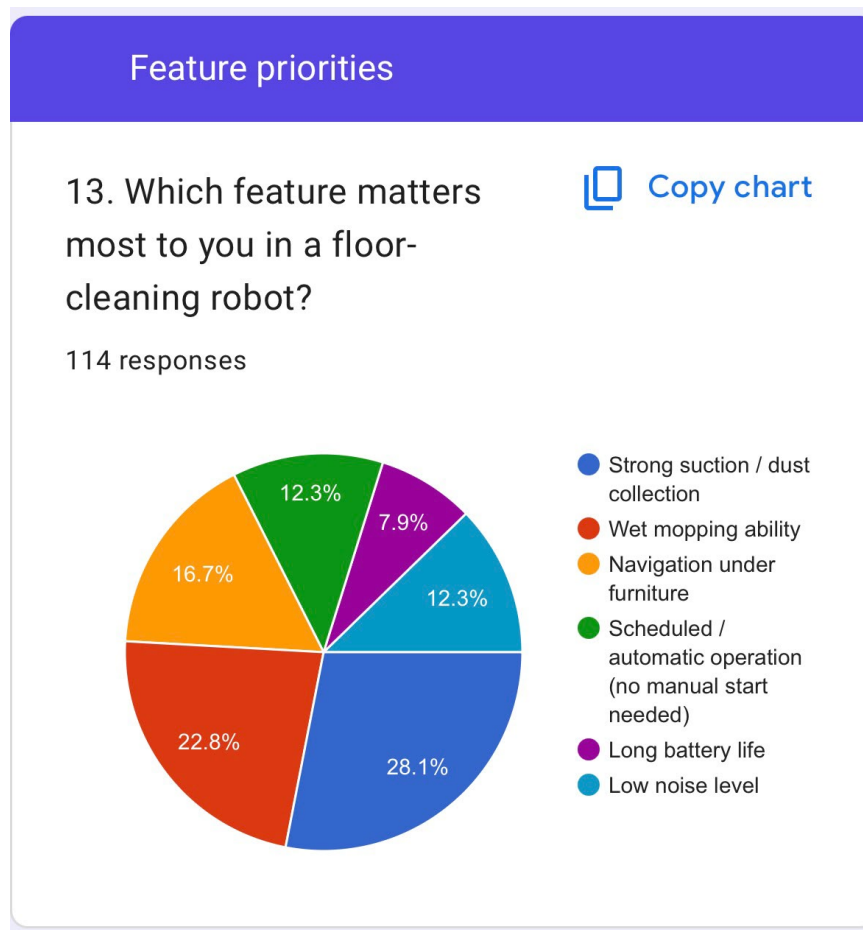


Figure 1.3: Which feature matters most to respondents in a floor-cleaning robot.

Strong suction/dust collection was the top priority at 28.1%, followed by wet mopping ability (22.8%) and navigation under furniture (16.7%). Scheduled/automatic operation and low noise level tied at 12.3% each, with long battery life the lowest priority at 7.9%. This ranking directly shaped several design decisions: it justifies prioritising suction performance and a robust brush/vacuum path in the current build (Section 3.2), motivates listing an active mopping module as the top item in the future improvements roadmap (Chapter 11) ahead of wet mopping being a launch feature, and confirms that investing in LiDAR-SLAM navigation (Section 3.3) is well justified given how highly respondents value reliable navigation under furniture. The comparatively lower priority placed on battery life suggests the $\sim 1.5\text{--}2$ hour runtime estimated in Section 6.2 is acceptable to the target market, provided suction and mopping performance are strong.

1.5 Justification for Project Selection

- Clear, survey-validated demand for an affordable autonomous floor cleaner in the local market.
- Existing gap between unaffordable imported robots and functionally poor low-cost alternatives.
- Skills alignment: the project draws on electronic circuit design, embedded control, communi-

cation/IoT integration and mechanical design, matching the team's engineering coursework.

- Scalable engineering scope: the project can be delivered as a working prototype within the semester timeline while leaving clear paths for future improvement (Chapter 11).

2. Technical Feasibility

This section shows that the proposed design can be built using tools, parts, and skills that are reasonably available to the team, and that the target performance is realistic.

2.1 Resource and Component Feasibility

Subsystem	Proposed Implementation	Feasibility Justification
Autonomous navigation	2D 360° LiDAR + Raspberry Pi 5 running ROS	Low-cost LiDAR sensors and ROS navigation tools (SLAM Toolbox, Nav2) are freely available and well documented online, so the team can build mapping and path-planning without designing new algorithms from scratch.
Motor control & drive	12V GA25-370 gear motors + DRV8251A motor driver	These motors and driver ICs are common, cheap, and easy to buy locally or import. They give enough torque to move a small robot chassis smoothly on tile or cement floors.
Suction & cleaning system	Centrifugal blower (PLA volute casing + impeller) driven by A2212 1400KV BLDC motor	The blower parts are 3D printed in-house using FDM, so no expensive tooling or outsourcing is needed. The team has already designed, printed, and tested a working prototype, proving the suction mechanism is achievable with current skills and equipment.
Power & battery life	4S LiPo pack + custom power PCB (buck + LDO regulation)	The power budget shows about 1 hour of runtime using a standard 4S 2200 mAh battery, which is enough for cleaning small indoor spaces and matches typical robot vacuum usage.
Manufacturing & fabrication	FDM 3D printing (PETG)	PETG provides good mechanical strength, impact resistance, and temperature stability while allowing rapid, low-cost fabrication using a standard FDM 3D printer.

Subsystem	Proposed Implementation	Feasibility Justification
Connectivity / mobile app	Wi-Fi + Bluetooth (via Raspberry Pi) + companion mobile app	Standard Wi-Fi/Bluetooth control methods (MQTT, REST, or BLE) are supported by many free libraries. Building a simple app with Flutter or React Native is realistic within the project timeline, since these tools are beginner-friendly and widely used.

2.2 Performance Targets

- Obstacle detection and avoidance response time: <200 ms (achievable at typical STM32F7 control-loop rates of 100 Hz+).
- Cliff-fall prevention: IR-based cliff sensors with a detection threshold tuned to typical furniture-leg/step heights found in local homes/offices.
- Target floor coverage: single-charge run sufficient to clean a small room/office ($\sim 20\text{--}30\text{ m}^2$) based on the power budget in Section 6.2.
- Navigation accuracy: LiDAR-SLAM localisation error target of a few centimetres, consistent with published performance of similar 2D LiDAR + ROS Nav2 setups on comparable hardware.

2.3 Key Technical Risks

Risk	Impact	Mitigation
Raspberry Pi 5 + LiDAR SLAM computational load vs. battery draw	Reduced runtime	Profile CPU/GPU load early; fall back to a lighter SLAM front-end if needed; separate power rail for RPi with adequate headroom.
Two-board (Power PCB + Controller PCB) interconnect reliability	Intermittent faults, brownouts under motor stall current	Use adequately rated connectors/wire gauge, add bulk capacitance at the Controller PCB input, fuse/protect the power rail.

3. Product Architecture

3.1 System Block Diagram

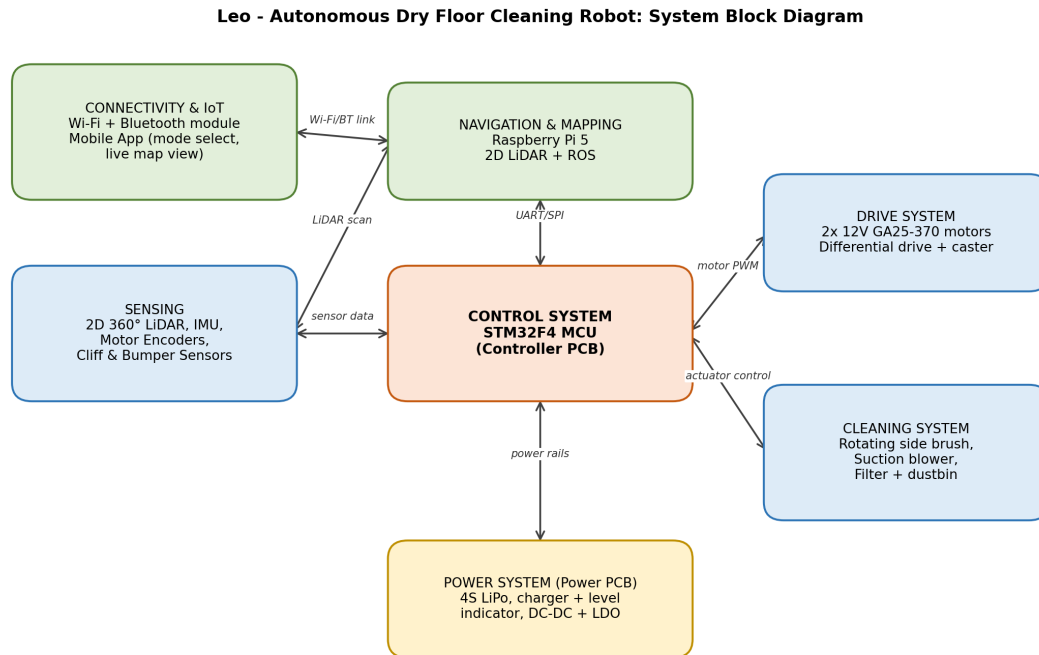


Figure 3.1: Leo system block diagram.

3.2 Block Descriptions

Block	Components	Function
Sensing	2D 360° LiDAR, IMU, motor encoders, cliff sensors, bumper switches	Provides environmental awareness for SLAM-based navigation, orientation feedback and safety cut-offs (cliff/collision).
Navigation & Mapping	Raspberry Pi 5 running ROS (SLAM Toolbox / Nav2)	Builds a live map of the environment from LiDAR scans, plans paths and issues high-level motion commands to the Controller PCB.
Control System	STM32F7 MCU on the Controller PCB	Executes the low-level control loop: motor PWM, encoder feedback, sensor polling, and interfaces with the Raspberry Pi over UART/SPI.
Drive System	2x 12V GA25-370 gear motors, differential drive, caster wheel	Provides locomotion and turning via differential-drive kinematics.
Cleaning System	Rotating side brush, suction blower, filter, removable dustbin	Sweeps debris from edges into the intake path and vacuums dust/hair into a removable, filtered dustbin.

Block	Components	Function
Power System	4S LiPo battery, Power PCB (charger, protection, buck/LDO regulation)	Supplies and regulates power to all subsystems and provides charge/level indication and protection.
Connectivity & IoT	Wi-Fi + Bluetooth (via Raspberry Pi), companion mobile app	Lets the user select cleaning modes and view the live generated map/status remotely.

3.3 Deviations from the Original Plan

The design evolved between the initial concept stage and the current architecture. The table below records the key deviations and the reasoning behind each change, as required for transparency in this report.

Area	Original Plan	Current / Final Approach	Reason for Change
Navigation strategy	Simplified reactive navigation (obstacle/cliff/bumper only, no complex mapping) as stated in the initial proposal	Full 2D LiDAR-SLAM navigation using ROS on a Raspberry Pi 5, with live map generation exposed in the mobile app	Reactive-only navigation gave poor coverage efficiency in early testing/simulation discussions; LiDAR + ROS was judged achievable within budget and gives a materially better user experience (map view, mode selection) that better matches survey-reported feature interest.
Compute platform	Raspberry Pi 5 or NVIDIA Jetson Nano (undecided) for computational tasks	Raspberry Pi 5 selected	Raspberry Pi 5 offers sufficient CPU headroom for the SLAM/navigation stack at lower cost and power draw than Jetson Nano-class boards, and has better local availability/support.

Area	Original Plan	Current / Final Approach	Reason for Change
Electronics architecture	Single custom PCB integrating MCU, motor drivers, sensing and power regulation	Split into two boards: a Controller PCB (STM32F7, sensor/motor-driver interfacing) and a separate Power PCB (charging, protection, DC-DC/LDO regulation)	Separating high-current power regulation from the sensitive digital control section reduces noise coupling into the MCU/sensor lines and simplifies debugging and independent replacement of either board.
App/mode control	Basic status indicators and control buttons only	Full companion mobile app with mode selection and live map generation view, over Wi-Fi/Bluetooth	Directly motivated by strong survey interest in app-based control and by the added value LiDAR mapping data can provide once collected.

4. PCB Design

The electronic system is split across two custom-designed printed circuit boards, both labelled **LEOv1**: a **Controller PCB** carrying the STM32F7 and drive/brush motor interfacing, and a **Power PCB** carrying battery management, charging, and voltage regulation. This separation isolates high-current switching/power circuitry from the sensitive digital control and sensor circuitry, improving noise immunity and allowing each board to be tested and replaced independently.

4.1 Controller PCB

The Controller PCB is built around an STM32F7 microcontroller (center, QFP package) and hosts the connectors and drive electronics required for navigation and cleaning actuation:

- **PWR / VMOTOR** – 12 V and 5 V power inputs from the Power PCB, plus a dedicated PWM + GND header for the main drive/motor supply.
- **LDM / RDM headers** – left and right drive motor connectors, each with ENA, ENB, 3V3, G, and – pins driving the on-board dual motor driver ICs.
- **RSBM / LSBM headers** – right and left side-brush motor outputs.
- **SIR1–SIR4** – four IR/line-following or cliff-sensor inputs.
- **MSW** – mechanical microswitch input (e.g. bumper contact).
- **BMI2C** – I²C header (SC, SD, GND) for the onboard IMU.

- **RPI / UART** – serial link headers to the Raspberry Pi 5 for high-level navigation commands.
- **DEBUG / USB** – SWD programming header (SW0, SCK, SDIO, NRST, 3V3) and a USB-C connector for flashing/debugging.
- **NRST / BOOT0** – reset push-button and boot-mode selector.

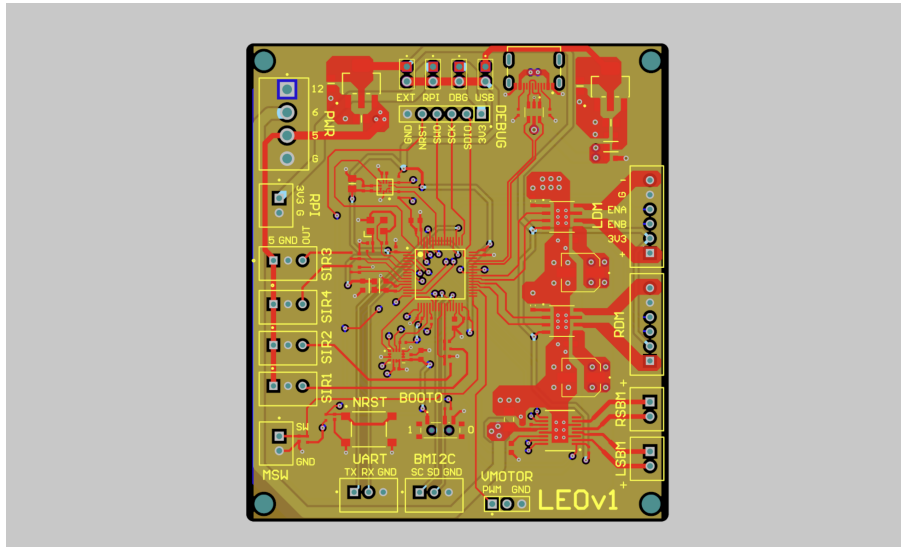


Figure 4.1: Controller PCB routed layout.

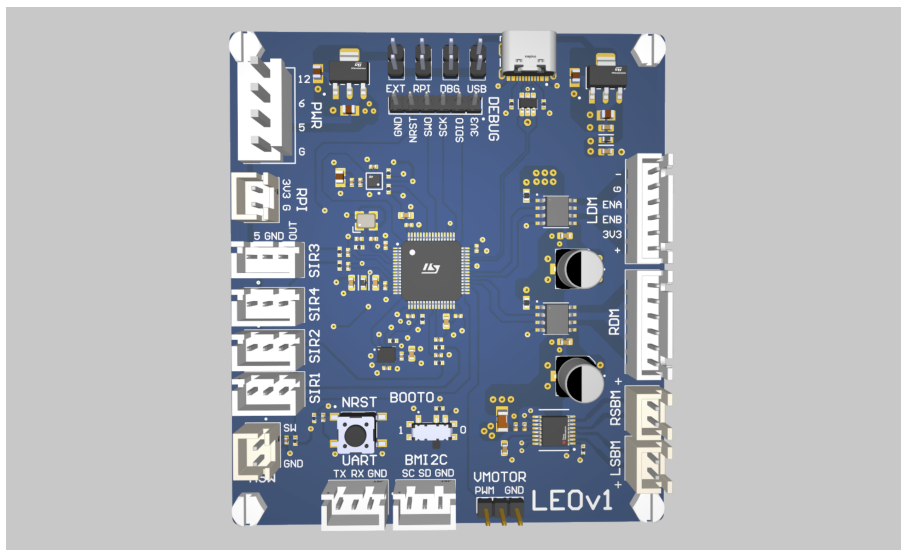


Figure 4.2: Controller PCB 3D render.

4.2 Power PCB

The Power PCB manages the 4S LiPo pack and distributes regulated rails to the rest of the system:

- **BATTERY / BAT CHA** – main pack connector and dedicated charging input jack.
- **TS1–TS3 / BALANCE PORT** – per-cell thermistor sense points and a balance-lead connector for safe multi-cell charging.

- **SENSE RES** – current-sense resistor array for over-current/coulomb-counting protection.
- **Buck converters** – two inductor-based switching regulators (top-right) step the battery voltage down to the 12 V and 5 V rails.
- **BLDC OUT** – dedicated high-current connector for the brushless suction blower motor.
- **RPI OUTS** – regulated 5 V USB-C output feeding the Raspberry Pi 5.
- **BAT12C / DEBUG / SW / NRST / BOOT0** – I²C link to a battery fuel-gauge/protection IC, SWD programming header, and boot/reset controls, mirroring the Controller PCB’s debug interface.

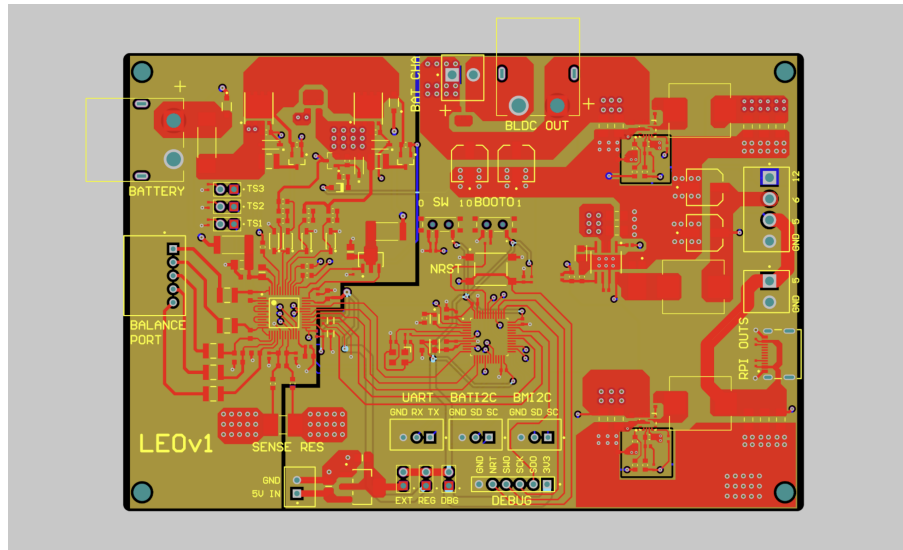


Figure 4.3: Power PCB routed layout.

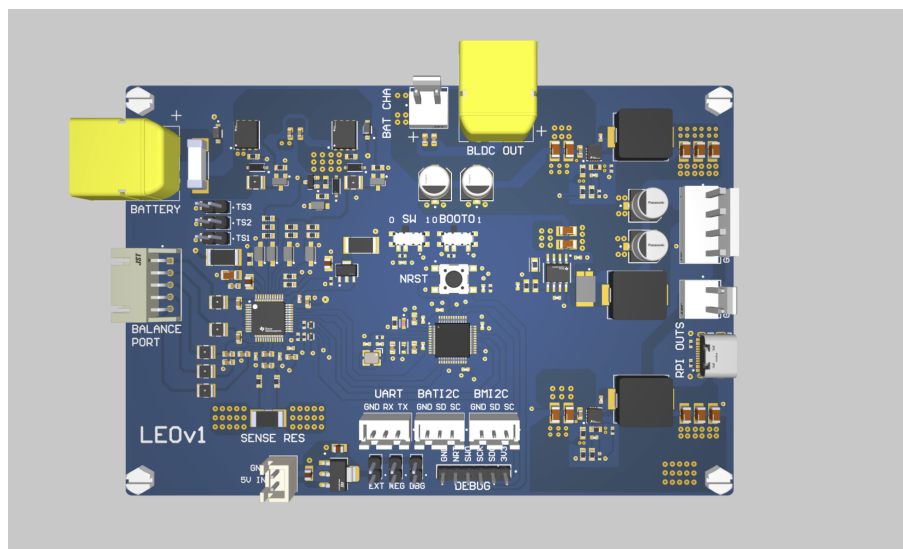


Figure 4.4: Power PCB 3D render.

4.3 PCB Stack-up and Justification

Both boards are specified as **4-layer, 1.6 mm FR-4** boards, fabricated by JLCPCB using their standard layer stack. The two dedicated internal planes give a solid, low-impedance ground

return and a clean power plane for the 12 V/5 V/3.3 V rails, which reduces EMI and noise coupling into the sensor and MCU signal traces compared to a 2-layer design – important given the switching buck converters on the Power PCB and the motor driver switching noise on the Controller PCB. Routing density around the STM32F7 QFP and the dual motor drivers is also significantly easier with dedicated inner planes, avoiding long ground-return loops. Copper weight is 1 oz on all layers for the Controller PCB; the Power PCB uses 2 oz copper on the outer layers carrying battery and motor current to limit trace heating and voltage drop. JL-CPCB's standard 4-layer service was chosen over 2-layer for the marginal cost increase, given the EMI and routing benefits it provides for a board handling both switching power regulation and sensitive digital signals.

5. Enclosure Design – Initial and Final Sketches

5.1 Initial Concept Sketch

The initial concept stage captured the overall subsystem breakdown as a hand-drawn mind map covering sensing, drive, cleaning, control, power, connectivity and navigation, used to scope the project before formal architecture work began.

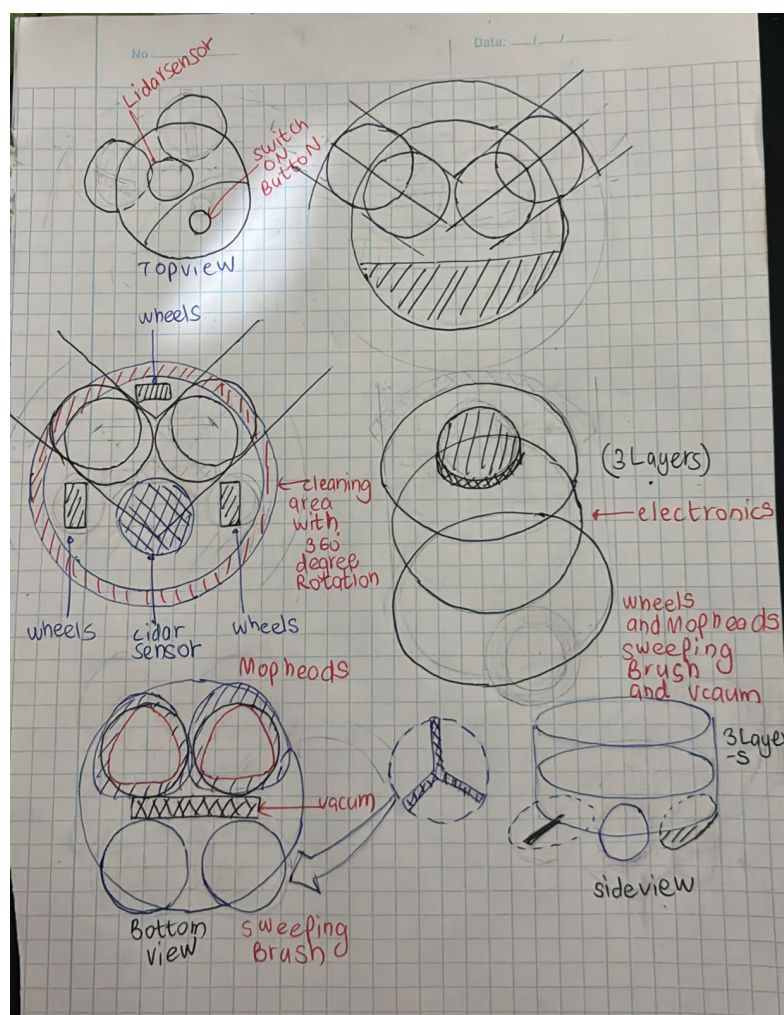


Figure 5.1: Initial concept

5.2 Final Enclosure Design

The enclosure forms the main body of the cleaning robot that protects all the internal mechanisms and electronics. This body houses all the important components including the battery, printed circuit board (PCB), drive motors, vacuum system, cables, and others, keeping them safe from dust, small impacts, and other environmental factors. In addition to protecting the internal elements, the enclosure provides mounting places for each of the subsystems, thus increasing the reliability of the robot.

The enclosure was created to meet all the functional and production needs. Considering that the total robot diameter equals 350 mm, the enclosure was created with the idea of maximizing the internal volume while creating the compact and low-volume robot body. The sufficient volume was provided for the battery, PCB, vacuum system, drive motors, and cable management. Moreover, the mounting places were provided for all important elements making it possible to mount them correctly and distribute the weight evenly.

Material Selection and Manufacturing Process

The enclosure was fabricated using the 3D printing technique known as Fused Deposition Modelling (FDM) with Polyethylene Terephthalate Glycol (PETG) filament. The use of PETG is motivated by the fact that it gives a good combination of mechanical properties along with being easy to print. Unlike PLA filament, PETG is stronger, has better impact resistance, and will last much longer. It is important for our case since the machine can be subjected to multiple impacts during the cleaning process.

Assembly Method

The box comprises upper and lower compartments separately that are joined together with standard M3 screws. These screws make it easy to take apart the robot for maintenance and repairs. Not only do the screws ensure that the compartments are firmly fixed mechanically but they also avoid the use of any glue that would make it difficult to change any damaged parts without having to replace the whole unit.

Internal Layout and Mounting Plates

The internal layout of the enclosure was designed to accommodate the robot's subsystems efficiently. The LiDAR sensor is positioned at the front of the robot, providing a clear field of view for obstacle detection and navigation. The battery compartment is located to maintain a low centre of gravity, improving the robot's stability during movement. Dedicated mounting locations are also incorporated for the drive motors, vacuum fan, caster wheel, and cleaning mechanism, while cable routing paths help organise wiring and reduce interference with moving components.

Beyond the main shell, the design uses two dedicated internal mounting plates rather than mounting components directly onto the main enclosure:

- **Motor mounting plate** – a plate that carries the two drive motors and associated hardware, positioned symmetrically for the differential-drive layout (Figure 5.3).
- **PCB / Raspberry Pi mounting plate** – a separate plate that carries the Controller PCB and the Raspberry Pi, mounted using spacers to provide sufficient clearance for wiring underneath and to minimise the transmission of vibration from the chassis into the electronics (Figure 5.4).

Separating these two functions onto their own plates keeps the electronics and the drivetrain mechanically decoupled, so either subsystem can be removed, serviced, or replaced without disturbing the other, and vibration from the motors is not transmitted directly into the PCB mounting points.

Overall, the enclosure provides a lightweight, durable, and modular structure that protects the robot's internal systems while supporting reliable operation and straightforward maintenance. The use of PETG as the printing material, combined with an FDM manufacturing process and M3 screw-based assembly, results in an enclosure that is practical to manufacture, sufficiently robust for the intended application, and suitable for iterative design improvements during product development.

Enclosure Views

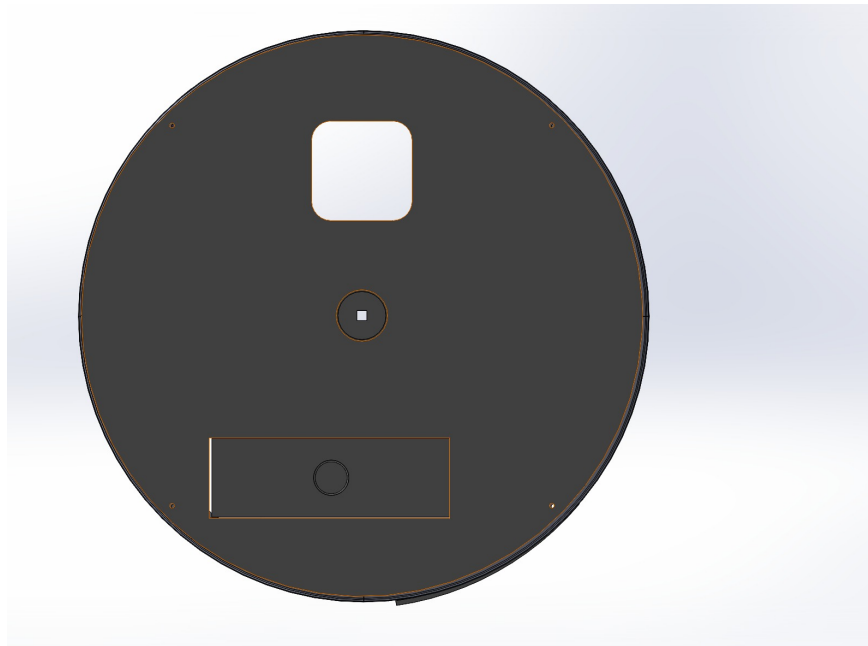


Figure 5.2: Top (lid) view – LiDAR aperture (square cutout), central cable pass-through, and the vacuum outlet slot with a circular port.

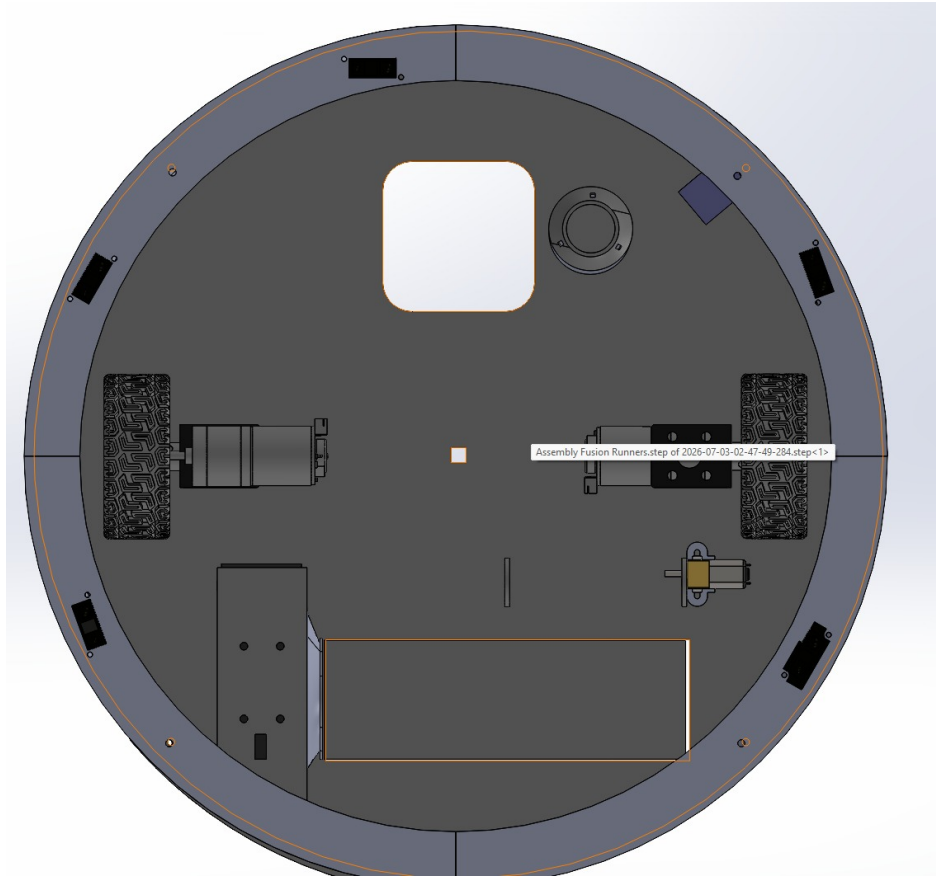


Figure 5.3: Bottom shell (internal view) – LiDAR mounting cutout, drive motor pockets either side, caster wheel mount, vacuum unit bay, and battery compartment.



Figure 5.4: Side view – low-profile chassis showing the drive wheel, caster wheel, and the ventilated vacuum-unit housing.

Motor and PCB Mounting Plates

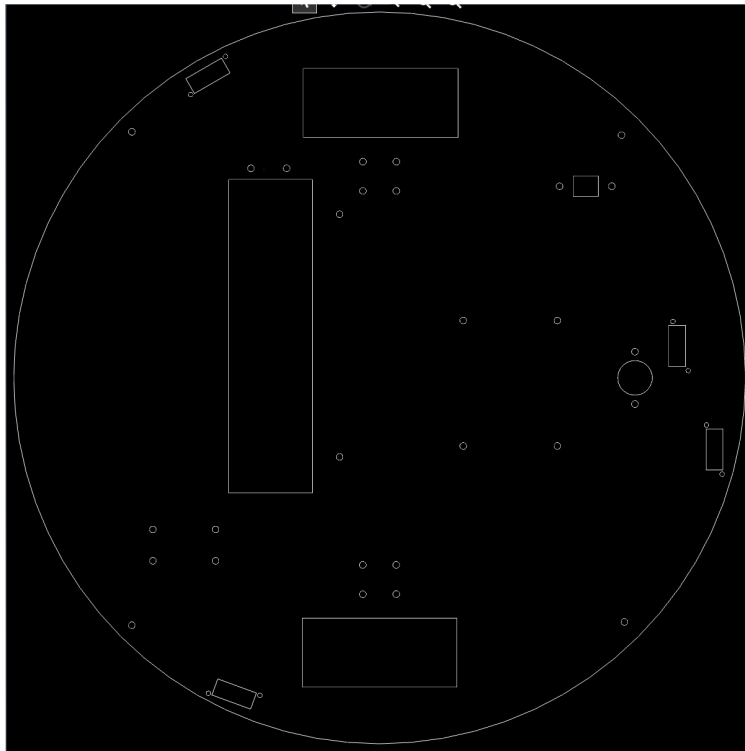


Figure 5.5: Motor mounting plate (DXF profile) – cutouts and fastener holes for the two drive motors, caster post, and vacuum unit bracket.

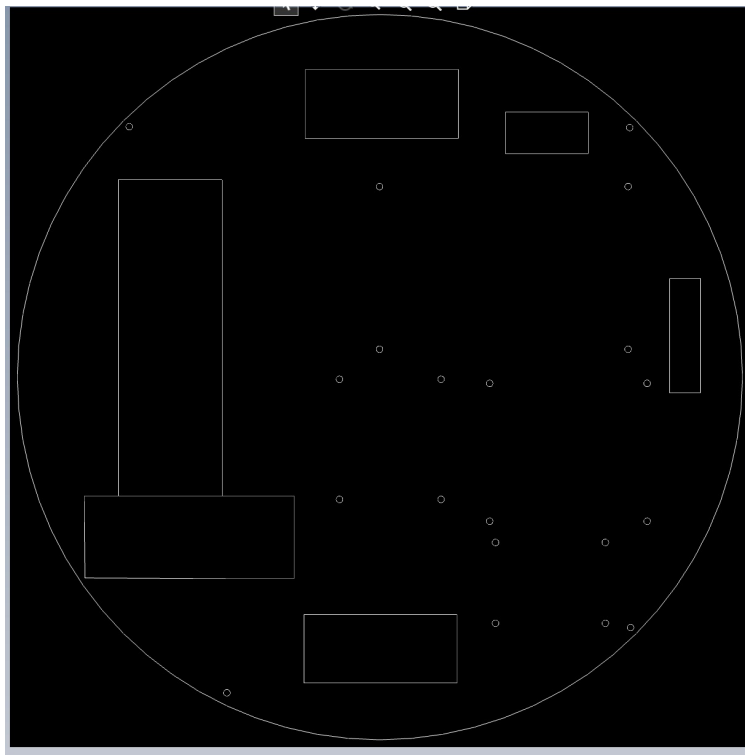


Figure 5.6: PCB / Raspberry Pi mounting plate (DXF profile) – standoff hole pattern for the Controller PCB, Power PCB, and Raspberry Pi 5, with a central cutout for cable routing.

The design path moved from a functional subsystem breakdown (Figure 5.1) toward this com-

pact, modular circular chassis, a form factor chosen because it simplifies rotational obstacle-recovery manoeuvres, matches the mounting geometry required for a 360° LiDAR turret near the front of the robot, and allows the motor and electronics mounting plates to be serviced independently of the main shell.

5.3 Centrifugal Vacuum Blower Design and Fabrication

The floor cleaner uses a small centrifugal vacuum blower that was designed and produced to produce vacuum. The blower design comprises a volute housing, a back-curved impeller, a suction axial inlet, and a tangential outlet. Mechanical components of the blower were designed on CAD and then printed via FDM from PLA material.

The impeller is attached directly to an A2212 1400KV Brushless DC Motor. At the time of operation, when the impeller rotates, it creates a low-pressure area in its center (eye) and pulls the air inside the blower through the axial inlet. The air takes kinetic energy and moves radially outwards under the effect of centrifugal forces. The volute housing transforms some portion of the speed of air into pressure. Air is exhausted through the tangential exit due to the created pressure difference, which causes the suction.

To avoid shaft misalignment and vibration during high-speed rotation, a motor mounting structure was included in the blower housing design. Also, the small volute shape was chosen to decrease the size of the blower. PLA was chosen because of its good accuracy, light weight, easy production process, and low costs of prototype creation.

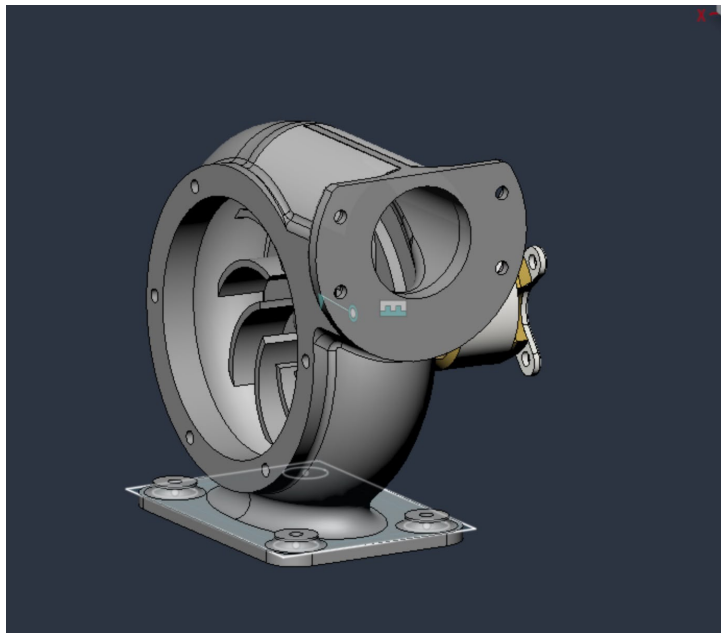


Figure 5.1: Isometric CAD view of the centrifugal vacuum blower assembly

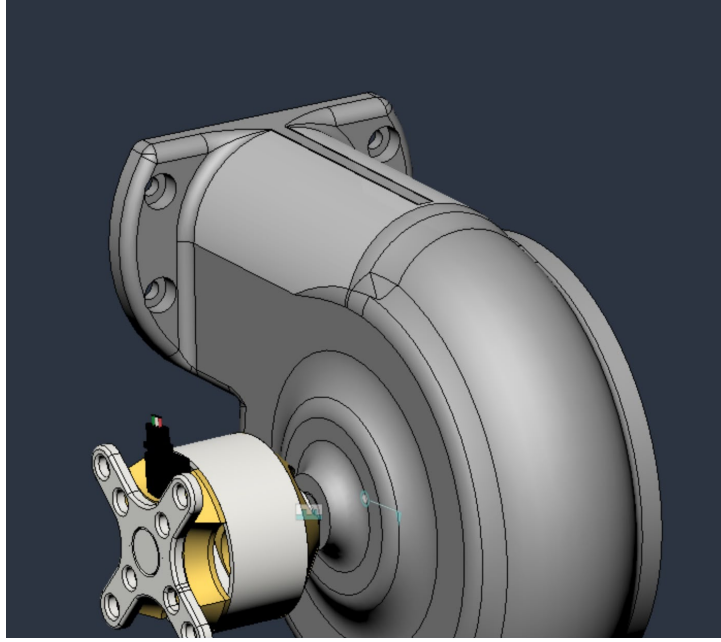


Figure 5.2: CAD view showing motor mounting arrangement and impeller coupling

5.3.1 Working Principle

The vacuum blower operates based on the centrifugal principle:

1. Air enters through the central inlet.
2. The high-speed impeller accelerates the air outward.
3. The volute casing collects the airflow and converts velocity into pressure.
4. Air exits through the tangential outlet, creating a low-pressure region at the inlet that generates continuous suction.

5.3.2 Key Specifications

Table 5.1: Vacuum Blower Specifications

Parameter	Specification
Blower Type	Centrifugal Vacuum Blower
Material	PLA (3D Printed)
Motor	A2212 1400KV BLDC Motor
Impeller Type	Backward-curved Centrifugal Impeller
Inlet	Axial (Center)
Outlet	Tangential
Drive Method	Direct Motor Coupling
Manufacturing Method	FDM 3D Printing
Application	Robot Vacuum Suction System

The developed blower provides a lightweight, compact, and cost-effective solution capable of generating continuous suction for dust collection in the robotic vacuum cleaner while demon-

strating the operating principle of centrifugal air movement.

6. Final Product Details

6.1 Product Specifications

Parameter	Value
Dimensions (Diameter × Height)	350 × 100 mm
Weight	[to be measured on final build] kg
Drive	Differential drive, 2x 12V GA25-370 motors + 1 caster wheel
Navigation	2D 360° LiDAR + IMU + wheel encoders, ROS SLAM on Raspberry Pi 5
Control MCU	STM32F7
Cleaning method	Rotating side brush + suction vacuuming, filtered dustbin
Connectivity	Wi-Fi + Bluetooth, companion mobile app
Battery	4S LiPo, 2200 mAh
Estimated average power draw	~40 W
Estimated runtime per charge	~60 minutes
Charging	On-board charge management with level indicator

6.2 Power Consumption and Battery Life Estimate

The table below gives an estimated power budget based on typical datasheet/operating current figures for each subsystem at full simultaneous operation (worst case: driving, cleaning and navigating at once). These are engineering estimates to guide battery sizing and should be validated with bench measurements once the prototype is assembled.

Subsystem	Voltage (V)	Avg. Current (A)	Power (W)
Drive motors (2x GA25-370)	12	1.5	18.0
Suction blower	12	0.8	9.6
Brush motor	6	0.3	3.6
2D LiDAR	5	0.4	2.0
STM32F7 + sensors + drivers (logic)	3.3	0.3	1.5
Wi-Fi / Bluetooth link	5	0.1	0.5
Total (average, all systems active)	–	–	35.2

With a 4S (14.8 V nominal) LiPo pack of 2200 mAh capacity, available energy is approximately

$14.8\text{ V} \times 2.2\text{ Ah} = 32.56\text{ Wh}$. Applying an 80% depth-of-discharge margin gives $\approx 26.05\text{ Wh}$ usable energy. At an estimated average draw of $\sim 35.2\text{ W}$, expected runtime is approximately $26.05\text{ Wh}/35.2\text{ W} \approx 0.74$ hours (≈ 44 minutes) per charge under continuous worst-case load – sufficient to clean a small room or office between charges, consistent with the performance target in Section 2.2.

6.3 Product Images



Figure 6.1: Final Product

7. Manufacturing Considerations for Volume Production

Moving from a single hand-assembled prototype to volume production requires attention to the following factors:

- **Design for Manufacture (DFM):** simplifying the enclosure for injection moulding (draft angles, uniform wall thickness, minimising undercuts) instead of 3D-printed prototype parts.
- **Design for Assembly (DFA):** reducing fastener count and using snap-fits/self-locating features to cut assembly time and labour cost per unit.
- **PCB panelisation:** panelising the Power PCB and Controller PCB for pick-and-place assembly (SMT) rather than hand-soldering, reducing per-unit assembly cost and improving consistency.
- **Component sourcing and second-sourcing:** qualifying at least one alternative supplier for key components (motors, LiDAR module, battery cells, MCU) to avoid single-supplier risk and long lead times.
- **Quality control and testing:** defining an end-of-line test procedure (motor current draw, sensor sanity checks, Wi-Fi pairing, battery charge/discharge check) that can be run quickly per unit on a production line.
- **Certification and compliance:** EMI/EMC and battery safety certification requirements (e.g. relevant Sri Lanka Standards Institution and/or international IEC/UN38.3 battery transport requirements) must be budgeted for before mass sale.
- **Supply chain and lead time:** LiDAR modules, LiPo cells and the Raspberry Pi compute module currently have the longest and most variable lead times among the BOM and should be ordered furthest in advance.
- **Firmware/software versioning:** a mechanism for field firmware updates (over Wi-Fi) should be designed in from the start to allow post-sale bug fixes without physical recall.

8. Product Costing (Volume Production) and Selling Price

8.1 Estimated Volume-Production Bill of Materials

Item	Qty	Est. Unit/Line Cost (LKR)
Chassis & enclosure (injection moulded, volume)	1	14000
2x GA25-370 gear motors + wheels	1 set	6000
Caster wheel	1	400

Item	Qty	Est. Unit/Line Cost (LKR)
2D 360° LiDAR module	1	20000
Cliff sensors (IR) x4 + bumper switches x2	1 set	4000
Raspberry Pi 5 (compute module)	1	30000
STM32F7 MCU (volume-priced)	1	1000
Motor driver ICs (drive + brush)	3	600
Suction blower motor	1	1500
Brush + filter assembly	1 set	700
4S LiPo battery pack (2200 mAh)	1	5800
BMS / protection module	1	450
Charging module	1	550
DC-DC converters + LDO regulators	1 set	500
Wi-Fi/BT (on Raspberry Pi – no extra cost)	–	0
Power PCB (fabricated + assembled, volume)	1	12500
Controller PCB (fabricated + assembled, volume)	1	7500
Wiring, connectors, others .	1 set	2000

Estimated total component cost per unit (volume pricing, quantities of ~500+ units): approximately **LKR 108,700**.

8.2 Selling Price Determination

Cost Element	Estimated Amount (LKR)
Component BOM (volume)	108,700
Manufacturing/assembly labour (per unit)	1,500
Packaging	1,000
Quality testing/certification (amortised per unit)	2,500
Total manufacturing cost per unit	113,700
Selling price (100% profit)	227,400

9. Marketing, Sales and Beyond

9.1 Target Market

- Middle-income households in urban and semi-urban Sri Lanka seeking an affordable alternative to imported robot vacuums.

- Boarding houses, small offices, and classrooms aiming to reduce recurring manual-cleaning labour and time.
- Local hobbyist and educational institutions interested in low-cost robotics platforms built using 3D-printed, locally sourced components.

9.2 Positioning

Leo is positioned as an affordable, locally-serviceable alternative to imported robot vacuums. Its centrifugal suction system, built using a 3D-printed volute casing and impeller driven by a low-cost BLDC motor, keeps manufacturing and repair costs significantly lower than imported units that rely on proprietary suction modules. Leo’s core value proposition is “80% of the functionality, at a fraction of the cost, with local support and repairability” – directly addressing the cost and after-sales service barriers identified in the survey.

9.3 Go-to-Market Approach

- Direct-to-consumer sales through a simple e-commerce storefront and social media platforms (Facebook/Instagram marketplace), which are widely used for product discovery in Sri Lanka.
- Partnerships with local electronics and appliance retailers to host physical demo units, allowing customers to see the suction performance first-hand.
- University and tech-fair showcases to demonstrate the 3D-printed centrifugal blower design, building credibility around Leo’s engineering and attracting early adopters.
- Referral and warranty-registration incentives through a companion mobile app, creating a direct customer channel for support, spare-part sales, and future upsells.

9.4 After-Sales and Beyond

- A local repair and service centre network, supported by the fact that Leo’s blower housing, impeller, and casing are 3D-printed and can be reprinted or repaired locally – a key differentiator versus imported products with sealed, non-serviceable suction units.
- Consumable spares (impellers, filters, brushes, BLDC motors, batteries) sold as accessories, creating a recurring revenue stream while leveraging Leo’s simple, modular blower design for easy part replacement.
- Firmware updates delivered over Wi-Fi to add features (e.g., scheduled cleaning, suction-mode adjustment, obstacle-avoidance improvements) post-purchase.
- Planned product line extension: a mopping-capable variant and a higher-suction model (using an upgraded BLDC motor and optimized volute design) for larger offices and commercial spaces, based on the roadmap in Chapter 11.

10. Project Budget with Bill of Quantities (BOQ)

Note (to complete): Prototype-stage prices below are estimated (LKR) for a single hand-built unit and are higher per-unit than the volume-production BOM in Chapter 8, which assumes

bulk purchasing. Confirm actual figures against your team's real purchase receipts.

Item	Qty	Unit Price (LKR)	Line Total (LKR)
Chassis/enclosure (prototype, 3D-printed)	1	20000	20000
2x GA25-370 gear motors + wheels	1 set	8000	8000
Caster wheel	1	450	450
2D 360° LiDAR module	1	30000	30000
Cliff sensors (IR) x4 + bumper switches x2	1 set	4000	4000
Raspberry Pi 5 (8GB)	1	60000	60000
STM32F7 MCU	1	1800	1800
Motor driver modules (drive + brush + blower)	3	1000	3000
Suction blower motor	1	4000	4000
Brush + filter assembly (prototype, sourced/adapted)	1 set	2,000	2,000
4S LiPo battery pack (2200 mAh)	1	6,500	6,500
Charging module	1	800	800
DC-DC converters + LDO regulators	1 set	600	600
Power PCB fabrication + assembly (prototype run)	5	15000	75000
Controller PCB fabrication + assembly (prototype run)	5	9000	36000
Wiring, other.	1 set	2,500	2,500

Estimated total prototype budget: approximately **LKR 253,600**.

11. Future Improvements

- Add an active mopping module (water tank + microfiber pad) to complete the sweep-vacuum-mop cycle referenced in the original proposal.
- Multi-floor mapping and persistent maps, allowing the robot to remember room layouts across cleaning sessions.
- Scheduled/automatic cleaning via the mobile app, with geofenced no-go zones.
- Migrate high-current motor driver stages to a more efficient/compact driver IC to extend battery life.
- Voice-assistant integration (e.g. via the mobile app) for hands-free control.

- Ingress protection improvements (dust/liquid sealing) for the underside of the chassis to improve durability.

12. Task Allocation Among Group Members

Team Member	Primary Responsibility
Fernando LTJ (230183R)	PCB Design, Schemetic Design, Autonomous Navigation using ROS, Component Selection, Prototype Testing, Assembly
Pathirana PTS (230468V)	PCB Design, Scemetic Design, Centrifugal Pump Case Design by Solidworks, Testing and Debugging, Component Selection, Assembly
Prishmika HWN (230500N)	PCB Design, Schemetic Design, Prototype Testing, Centrifugal Pump Impeller Design, Testing and Debugging, Assembly
Withanage GDN (230735M)	Complete Encloser Design with each and every part using solid works and Assembly Roller Brush and Rotating Brush Design Testing and Debugging, Designed the robot's mechanical system, including the chassis, battery holder, brush assembly, airflow path, and component integration while ensuring manufacturability and structural reliability
Hendeniya HMJC (230238R)	Schemetic Design, Develop a Mobile App, Testing and Debugging, Documentation